Data period

1. The dataset spans a 3-month collection period, ranging from January 1, 2016 to March 31, 2016.

   Syntax used:

```
-- They started collecting data on  2016-01-01
SELECT MIN(RecordDate2) AS  Min_date
FROM workspace.default.viewership;

-- They last collected data on 2016-03-31
SELECT MAX(RecordDate2) AS Max_date
FROM workspace.default.viewership;

-- The duration on the data is 3 months
```

Min_date
2016-01-01T00:32:00.000+00:...

Max_date
2016-03-31T23:37:00.000+00:...

Distinct Shows Broadcasted

2. Bright TV's content library comprises 21 distinct shows currently airing across its broadcasting channels.

   Syntax used:

```
SELECT DISTINCT Channel2
FROM workspace.default.viewership;

SELECT COUNT(DISTINCT Channel2) AS Number_of_Shows
FROM workspace.default.viewership;
```

Tables of shows :

	Channel2
1	Trace TV
2	ICC Cricket World Cup 2011
3	SawSee
4	Boomerang
5	Channel O
6	Cartoon Network
7	Supersport Live Events
8	SuperSport Blitz
9	E! Entertainment
10	Break in transmission
11	kykNET
12	Africa Magic
13	CNN
14	Vuzu
15	M-Net
16	DStv Events 1
17	MK
18	SuperSport Live Events
19	Live on SuperSport
20	Sawsee
21	Wimbledon

Combined User Profile Table and Viewership Table

Joined the two tables (Userprofile and Viewership) on a full outer join as I believe there are insights that can also be made from users who have profiles but have not watched any shows in the sample period.

It is also important to note that I did not include each users name, surname and email addresses as I do not believe that it will provide any significant insight to know the users personal details.

Syntax used:

```sql
SELECT
    -- User Profile columns
    up.UserID        AS up_UserID,
    up.Gender,
    up.Race,
    up.Age,
    up.Province,

    -- Viewership columns
    v.UserID0         AS v_UserID,
    v.Channel2,
    v.RecordDate2,
    v.`Duration 2`

FROM workspace.default.userprofiles AS up
FULL OUTER JOIN workspace.default.viewership AS v
    ON up.UserID = v.UserID0;
```

Table +

	p_UserID	Gender	Race	Age	Province	v_UserID	Channel2	RecordDate2	Duration 2
1	808923	male	other	23	Eastern Cape	808923	SuperSport Blitz	2016-02-21T18:03:00.000+00:...	1899-12-31T00:01:1
2	808923	male	other	23	Eastern Cape	808923	Trace TV	2016-02-17T18:45:00.000+00:...	1899-12-31T00:13:2
3	630977	male	coloured	28	Gauteng	630977	Trace TV	2016-01-16T09:52:00.000+00:...	1899-12-31T00:00:1
4	850656	male	black	26	Mpumalanga	850656	Africa Magic	2016-03-05T21:12:00.000+00:...	1899-12-31T00:00:1
5	850656	male	black	26	Mpumalanga	850656	Channel O	2016-03-06T05:25:00.000+00:...	1899-12-31T00:00:0
6	850656	male	black	26	Mpumalanga	850656	Africa Magic	2016-03-06T07:09:00.000+00:...	1899-12-31T00:00:0
7	2368365	male	coloured	27	Limpopo	2368365	ICC Cricket World Cup 2011	2016-03-30T11:17:00.000+00:...	1899-12-31T00:38:1
8	1106115	female	black	26	Western Cape	1106115	Africa Magic	2016-01-11T16:06:00.000+00:...	1899-12-31T00:06:2
9	1106115	female	black	26	Western Cape	1106115	Trace TV	2016-01-11T15:33:00.000+00:...	1899-12-31T00:05:5
10	820387	male	black	15	Limpopo	820387	SuperSport Blitz	2016-01-27T13:54:00.000+00:...	1899-12-31T00:12:0
11	820387	male	black	15	Limpopo	820387	Cartoon Network	2016-01-27T09:52:00.000+00:...	1899-12-31T00:01:2
12	820387	male	black	15	Limpopo	820387	Channel O	2016-01-25T10:07:00.000+00:...	1899-12-31T00:00:5
13	820387	male	black	15	Limpopo	820387	SuperSport Blitz	2016-01-26T19:30:00.000+00:...	1899-12-31T00:07:3
14	808984	male	coloured	42	Western Cape	808984	kykNET	2016-02-19T15:58:00.000+00:...	1899-12-31T00:01:5
15	778615	None	None	0	None	778615	Channel O	2016-03-12T15:45:00.000+00:...	1899-12-31T00:03:4
16									

10,989 rows | 35.93s runtime See performance (1) Optimize Refreshed now

Cleaning Dataset with Null Functions

To ensure data integrity and consistency across both tables, null values were addressed using the IFNULL function, which replaces any missing or null entries with a meaningful default value rather than leaving them blank.

Syntax used:

```
SELECT
    -- User Profile columns & handling null values

    IFNULL(CAST(up.UserID   AS STRING), '0')              AS up_UserID,
    IFNULL(up.Gender,                   'No_Gender')      AS Gender,
    IFNULL(up.Race,                     'No_Race')        AS Race,
    IFNULL(CAST(up.Age      AS STRING), 'No_Age')         AS Age,
    IFNULL(up.Province,                 'No_Province')    AS Province,

    -- Viewership columns & handling null values

    IFNULL(CAST(v.UserID0       AS STRING), '0')          AS v_UserID,
    IFNULL(v.Channel2,                      'No_Channel')  AS Channel2,
    IFNULL(CAST(v.RecordDate2   AS STRING), 'No_RecordDate')AS RecordDate2,
    IFNULL(CAST(v.`Duration 2`  AS STRING), '0')          AS `Duration 2`,
```

Table

	up_UserID	Gender	Race	Age	Province	v_UserID	Channel2	RecordDate2	Duration 2
1	808923	male	other	23	Eastern Cape	808923	SuperSport Blitz	2016-02-21 18:03:00	1899-12-31 00:01:12
2	808923	male	other	23	Eastern Cape	808923	Trace TV	2016-02-17 18:45:00	1899-12-31 00:13:21
3	630977	male	coloured	28	Gauteng	630977	Trace TV	2016-01-16 09:52:00	1899-12-31 00:00:15
4	850656	male	black	26	Mpumalanga	850656	Africa Magic	2016-03-05 21:12:00	1899-12-31 00:00:1...
5	850656	male	black	26	Mpumalanga	850656	Channel O	2016-03-06 05:25:00	1899-12-31 00:00:00
6	850656	male	black	26	Mpumalanga	850656	Africa Magic	2016-03-06 07:09:00	1899-12-31 00:00:00
7	2368365	male	coloured	27	Limpopo	2368365	ICC Cricket World Cup 2011	2016-03-30 11:17:00	1899-12-31 00:38:12
8	1106115	female	black	26	Western Cape	1106115	Africa Magic	2016-01-11 16:06:00	1899-12-31 00:06:2...
9	1106115	female	black	26	Western Cape	1106115	Trace TV	2016-01-11 15:33:00	1899-12-31 00:05:57
10	820387	male	black	15	Limpopo	820387	SuperSport Blitz	2016-01-27 13:54:00	1899-12-31 00:12:08
11	820387	male	black	15	Limpopo	820387	Cartoon Network	2016-01-27 09:52:00	1899-12-31 00:01:27
12	820387	male	black	15	Limpopo	820387	Channel O	2016-01-25 10:07:00	1899-12-31 00:00:57
13	820387	male	black	15	Limpopo	820387	SuperSport Blitz	2016-01-26 19:30:00	1899-12-31 00:07:30

Create a time buckets

1. Age Buckets Rather than working with raw age values, users were segmented into six distinct life-stage groups:

- No_Age — for records where age is null
- Kid — users under 12
- Teenager — users aged 13 to 19
- Young Adult — users aged 20 to 34
- Matured Adult — users aged 35 to 59
- Senior Citizen — users aged 60 and above

Any age values that fall outside these ranges are labelled Unknown.

```sql
-- Age Buckets
CASE
    WHEN up.Age IS NULL                THEN 'No_Age'
    WHEN up.Age < 12                   THEN 'Kid'
    WHEN up.Age BETWEEN 13 AND 19      THEN 'Teenager'
    WHEN up.Age BETWEEN 20 AND 34      THEN 'Young Adult'
    WHEN up.Age BETWEEN 35 AND 59      THEN 'Matured Adult'
    WHEN up.Age >= 60                  THEN 'Senior Citizen'
    ELSE 'Unknown'
END AS Age_Group,
```

Table

	Age	Province	v_UserID	Channel2	RecordDate2	Duration 2	Age_Group	Day_Type	Day_Of_Week	Time_Of_Day
1		Eastern Cape	808923	SuperSport Blitz	2016-02-21 18:03:00	1899-12-31 00:01:12	Young Adult	Weekend	Sunday	Evening
2		Eastern Cape	808923	Trace TV	2016-02-17 18:45:00	1899-12-31 00:13:21	Young Adult	Weekday	Wednesday	Evening
3		Gauteng	630977	Trace TV	2016-01-16 09:52:00	1899-12-31 00:00:15	Young Adult	Weekend	Saturday	Morning
4		Mpumalanga	850656	Africa Magic	2016-03-05 21:12:00	1899-12-31 00:00:1...	Young Adult	Weekend	Saturday	Night
5		Mpumalanga	850656	Channel O	2016-03-06 05:25:00	1899-12-31 00:00:00	Young Adult	Weekend	Sunday	Morning
6		Mpumalanga	850656	Africa Magic	2016-03-06 07:09:00	1899-12-31 00:00:00	Young Adult	Weekend	Sunday	Morning
7		Limpopo	2368365	ICC Cricket World Cup 2011	2016-03-30 11:17:00	1899-12-31 00:38:12	Young Adult	Weekday	Wednesday	Morning
8		Western Cape	1106115	Africa Magic	2016-01-11 16:06:00	1899-12-31 00:06:2...	Young Adult	Weekday	Monday	Afternoon
9		Western Cape	1106115	Trace TV	2016-01-11 15:33:00	1899-12-31 00:05:57	Young Adult	Weekday	Monday	Afternoon